Assignment 2: Simple Harmonic motion (1)

Summer Break Bridge Course

Exercises

1. Force constant and its physical meaning

For the restoring-force law

$$F = -kx,$$

answer the following:

(i) What are the SI unit and the dimension of the force constant k?

(ii) If two springs are stretched by the same amount, but one has a larger value of k, which one exerts the larger restoring force?

(iii) If two particles have the same mass but are attached to springs with different k-values, which one will oscillate faster? Explain physically.

2. Solving SHM using an exponential trial solution

Consider the differential equation

$$\frac{d^2x}{dt^2} = -\omega^2 x.$$

(i) Assume a solution of the form

$$x(t) = e^{at}.$$

Substitute this into the differential equation and find 2 possible values of a.

(ii) You should be getting 2 values of a in terms of ω, let's say a_1 and a_2.Then 2 linearly independent solutions would be $e^{a_1 t}$ and $e^{a_2 t}$. Now, write the most general solution as a linear combination of these 2 solutions.

(iii) In your general solution, you would have 2 arbitrary constants. Can you get the physical meaning of these 2 constants? Are they in any way linked to the initial position and initial velocity? Get the exact relation.

(iv) Try to rewrite this solution in the form derived in class *i.e.* $x(t) = A\sin(\omega t)$. Use Euler relations,

$$e^{\pm i\theta} = \cos\theta \pm i\sin\theta.$$

3. A concrete SHM problem

At $t = 0$, a particle is at equilibrium and moving in the positive direction with speed 5 units. Its amplitude of oscillation is 15 units.

(i) Find its equation of motion in the form

$$x(t) = A\sin(\omega t + \alpha).$$

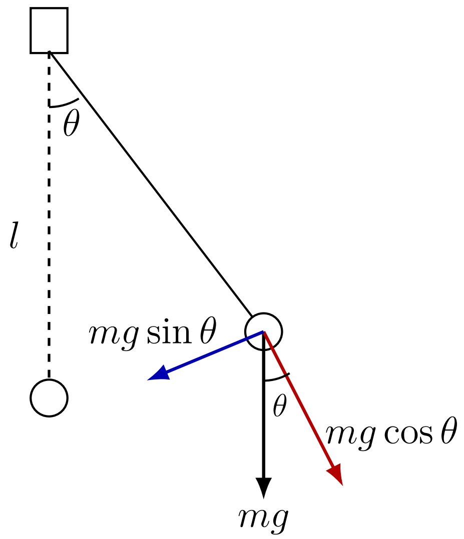
i.e. find the values of A and α.

(ii) If the force constant is $k = 3$ units and the mass is $m = 0.75$ units, find ω.

(iii) Plot $x(t)$ and $v(t)$ on the same graph in Python. Take the time array from $t = 0$ to $t = 5$ with 500 points. Important point: in NumPy, `sin` and `cos` take input in radians, not in degrees. So if you use the phase constant α, make sure its value is in radians.

4. Simple pendulum and SHM approximation

Consider a simple pendulum of length l and bob mass m, displaced by a small angle θ from the vertical.



(i) Resolve the weight mg into components and identify the restoring force.

(ii) Write the exact differential equation of motion.

(iii) For small θ, use the approximation $\sin\theta \approx \theta$ to simplify the equation.

(iv) Using $x = l\theta$, compare the result with

$$\frac{d^2x}{dt^2} = -\omega^2 x$$

and identify ω.

5. Damped oscillator

When friction acts on a particle in SHM, the total force has a friction-part that is proportional to velocity, together with the part proportional to displacement.

$$F = -b\,v - kx.$$

where v is its velocity and x is its displacement.

(i) Write the differential equation of motion in terms of x, $\frac{dx}{dt}$, and $\frac{d^2x}{dt^2}$.

(ii) Assume a solution of the form

$$x(t) = e^{rt}.$$

Substitute it into the differential equation and find the equation satisfied by r.

6. Phase-space relation

For SHM, show that

$$\frac{x^2}{A^2} + \frac{v^2}{A^2\omega^2} = 1.$$

What kind of curve does this represent in the x-v plane? *i.e.* x along x-axis and v along y-axis. Assume A and ω are constants.